

Maria Gratzia Indacochea Rosado

Software Engineer | Full Stack Developer

grtz2001@gmail.com · Jersey City, NJ · (612) 449-1874 · linkedin.com/in/gratzia · github.com/grtz2001

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with a strong Computer Science foundation (object oriented design, data structures, algorithms) specializing in Python web application and API development, backend integration, and scalable production platforms. Experienced building accessible, responsive interfaces from Figma using TypeScript and modern frameworks, and implementing unit & integration test suites and code review practices to meet engineering quality standards. Comfortable working with SQL/NoSQL databases, Java, Python, REST APIs, and prototyping AI-enabled workflows with LLMs; collaborative contributor across the full software development process.

EDUCATION

University of St. Thomas, B.S. Computer Science & B.A. Mathematics, *Summa Cum Laude* May 2024
Relevant coursework: Data Structures, Algorithms, Web Development, Artificial Intelligence & Robotics, Object Oriented Design

SKILLS

Database Technologies: SQL, MySQL, SQLite3, NoSQL, R, MATLAB, LaTeX, Query Optimization, Database Integration
Programming Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, C, C++, C#, PHP, JSON
Frameworks & Tools: Node.js, Vue.js, React.js, Vite, NextJS, .NET, Git, GitHub, Linux, GraphQL, LLMs, CRM systems
Competencies: Backend API Development, Data Analysis, Agile Collaboration, Responsive UI Design, WCAG 2.1 Compliance, Legacy Code Modernization, Internationalization, Automated Testing, Code Review, Unit Testing, UI/UX

PROFESSIONAL EXPERIENCE

Software Engineer, Full-Stack Developer Aug 2024 – Apr 2026
Edutrak Software Remote

- Rebuilt a legacy ESC user interface using TypeScript and modern frontend patterns, translating Figma mockups and product requirements into responsive, WCAG 2.1 AA-compliant layouts to improve usability and cross-device stability.
- Led migration from Vue 2 to Vue 3 with TypeScript and Pinia, consolidating multiple projects into a unified User Services platform and enabling single sign-on to streamline user workflows and reduce friction.
- Refactored Node.js REST APIs and database connectivity modules as part of system unification: implemented unit and integration test suites, documented REST endpoints and technical specifications, introduced code review practices, and improved debuggability to reduce regressions and shorten QA cycles while improving maintainability.
- Implemented localization by replacing hardcoded content with an i18n structure supporting EN, ES, and additional locales, enabling consistent translation management and easier expansion to new markets.
- Developed a dynamic form builder with a WYSIWYG editor, multi-page layouts, and responsive components; ensured WCAG 2.1 AA compliance and added backend unit testing coverage for critical form flows.

Teaching Assistant & Senior Math Tutor Sep 2022 – May 2024
Mathematics Dept., University of St. Thomas St. Paul, MN

- Distilled complex mathematical concepts (Pre-Calculus through Multivariable Calculus) into accessible explanations for diverse learners, adapting communication to individual needs to improve understanding and outcomes.
- Delivered constructive feedback and mentored 90+ students across three sections of Calculus I & II, applying attention to detail to ensure fairness and accuracy in evaluation and helping students improve problem-solving skills.

Researcher, CAM Summer Research Program Jun 2023 – Aug 2023
Center for Applied Mathematics (CAM), University of St. Thomas St. Paul, MN

- Served as primary student researcher on a Power Domination project, using graph theory to model power networks and optimize phasor measurement unit (PMU) placement.
- Designed and implemented algorithms to minimize the number of PMUs required to achieve full grid observability, evaluating algorithm performance and time/space complexity to improve efficiency and resource allocation.

ORGANIZATIONS & PROGRAMS

Ecuadorian Mathematical Organization (OMEC) | Volunteer Jul 2019 – Present
• Served as Deputy Leader of Ecuador's national team at EGMO (2020–2022, 2026 in Bordeaux, France); led selection, training, and coordination of top student talent for IMO and EGMO.

Pan-American Girls' Mathematical Olympiad (PAGMO) | Problem Coordinator 2021 – 2022
• Authored solutions and marking schemes, establishing consistent grading criteria for an international contestant pool.
• Conducted coordination meetings to review final scores and ensure consistency across evaluators.